

SOOYONG KIM

Robotics & Autonomous Systems Engineer

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PROFESSIONAL SUMMARY

Robotics and autonomous systems engineer with a strong mechanical-engineering research foundation in dynamics, mathematical modeling, state estimation, sensing, control, simulation, and experimental validation. Experienced in ROS/ROS 2, mobile robotics, sensor fusion, perception, path planning, and model-based robotics, with research experience in autonomous mobile robots at the University of Waterloo and vibration-based system identification and model validation at the University of Victoria. Current work focuses on analytical and simulation-based robotics development using Python, C++, ROS 2, and Isaac Sim, with emphasis on kinematics, dynamics, control, perception, and quantitative cross-validation. Strong focus on reconciling mathematical models, simulation behavior, and quantitative results to identify discrepancies in modeling assumptions, coordinate frames, interfaces, and implementation.

CORE TECHNOLOGIES

Robotics & Control: ROS 2, ROS, Nav2, SLAM, AMCL, RRT/RRT*, Pure Pursuit, MPC, manipulator kinematics, rigid-body dynamics

State Estimation & Fusion: EKF, IMU, wheel odometry, LiDAR-camera fusion, coordinate transforms, sensor calibration

Perception: OpenCV, YOLO, object tracking, instance/semantic segmentation, RGB-D perception

Programming & Simulation: Python, C++, MATLAB, Linux, Git/GitHub, Isaac Sim, Gazebo, TurtleBot, Summit XL, UR5e

SELECTED ROBOTICS PROJECTS

UR5e Modeling, Dynamics & Control

Python · C++ · ROS 2 · Isaac Sim · Rigid-Body Dynamics · Control

- Reconstructed a UR5e model from URDF-defined kinematic, inertial, and frame parameters and aligned analytical and simulated representations.
- Cross-validated poses, geometric Jacobians, joint-space mass matrices, and gravity terms against Isaac Sim with quantitative error analysis.
- Developing ROS 2/C++ manipulator control and trajectory tracking on the validated model, with RGB-D perception and manipulation experiments.

ROS 2 EKF Sensor Fusion

ROS 2 Humble · Gazebo · State Estimation

- Implemented wheel-odometry and IMU fusion with an Extended Kalman Filter and integrated simulated sensor streams and coordinate frames.
- Compared raw odometry, EKF estimates, and ground truth across curved and extended trajectories and quantified localization error.

Real-Time Vision Perception Pipeline

ROS 2 · PyTorch · Ultralytics YOLO · OpenCV · DeepSORT

- Built a reusable ROS 2 pipeline for object detection, instance segmentation, tracking, and quantitative performance logging.
- Achieved 31.95 FPS over 2,136 steady-state frames with 18.26 ms mean processing latency and 8.70 ms mean inference latency on RTX 5060 Laptop GPU.

LiDAR-Camera 3D Perception & Fusion

ROS 2 · Isaac Sim · LiDAR · RGB Camera · OpenCV · 3D Geometry

- Implemented calibrated LiDAR-to-camera projection and combined detection/segmentation with LiDAR returns for object-level distance estimation.
- Evaluated frame alignment, projection consistency, and depth robustness in Isaac Sim warehouse scenes.

EDUCATION

University of Waterloo

2020–2021

PhD Program, Mechanical & Mechatronics Engineering — Voluntary Withdrawal

University of Victoria

2016–2019

MASc, Mechanical Engineering — Thesis: Predicting Regenerative Chatter in Turning Using Operational Modal Analysis

Pusan National University

2005–2012

B.Sc., Mechanical Engineering — Undergraduate Thesis: Investigation of SOFC Injecting Natural Gas Directly

PROFESSIONAL EXPERIENCE

MKS Investments — Manager

2023 – 2026

Technology Research & Analysis / Business Development

- Reviewed and validated technical and market analyses, identifying incorrect assumptions, data inconsistencies, unsupported claims, and source-quality issues.
- Cross-checked technical feasibility, benchmarks, source data, and competitive evidence to improve decision reliability across emerging technologies.

- Managed technology/market research and translated technical findings into decision-oriented recommendations for management and external stakeholders.

RESEARCH EXPERIENCE

University of Waterloo — Research Assistant, Mechanical & Mechatronics Engineering

May 2020 – Dec 2021

Waterloo Mechanical Systems and Control Lab

- Conducted mobile-robotics research in autonomous navigation, localization, path planning, sensing, and physical robot experimentation using Summit platforms and motion-capture systems.
- Developed LiDAR-camera fusion for semantic road segmentation on KITTI and investigated transfer learning for robot semantic tasks.
- Implemented MPC-based mobile-robot navigation and performed physical experiments to evaluate path-following performance; supported motion-aware planning research.
- Developed and validated ROS/Gazebo laboratory material covering SLAM, AMCL, RRT/RRT*, Reeds-Shepp/Dubins planning, and Pure Pursuit tracking.

University of Victoria — Research Assistant, Mechanical Engineering

Aug 2017 – Apr 2019

Dynamics and Digital Manufacturing Lab

- Developed an Operational Modal Analysis method for estimating turning-process stability from in-process vibration measurements before regenerative chatter fully developed.
- Modeled machining dynamics using multi-DOF, state-space, and time-delay formulations; performed modal identification, stability analysis, and MATLAB-based signal processing.
- Designed instrumented machining experiments with accelerometers, NI DAQ, acoustic measurements, and experimental modal analysis; validated analytical and numerical predictions against physical experiments.
- Conducted laser Doppler vibrometry and output-only modal analysis for high-speed micromilling, comparing OMA with conventional impact-hammer EMA.

ADDITIONAL ENGINEERING & TEACHING

Solid Oxide Fuel Cell System Fueled Directly by Natural Gas

2011 – 2012

Pusan National University · Undergraduate Research

- Developed and experimentally evaluated a tubular SOFC system for direct natural-gas utilization; work led to a Korean patent application and university/external awards.

Teaching: Autonomous Mobile Robotics (University of Waterloo); Mechanical Vibrations, Mechanics of Solids II, and CAD-Aided Manufacturing (University of Victoria).

SELECTED PUBLICATION, PATENT & AWARDS

Publication: S. Kim and K. Ahmadi, "Estimation of vibration stability in turning using operational modal analysis," Mechanical Systems and Signal Processing, 2019.

Patent: Solid Oxide Fuel Cell System Fueled By Natural Gas Directly — Korean Patent Application No. 10-2012-0109237 (2012), inventor: Sooyong Kim et al.

Selected Awards: Grand Prize, Integrated Capstone Design Competition (2011); Grand Prize, Research Assessment of Industry-University Collaboration (2011); Excellence Award, Engineering Students Technology & Science Competition (2011).